

State Variable Theory

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#Explanation

Given our Euler-Lagrange Formulation:

$$B(q)\ddot{q} + C(\dot{q}, q)\dot{q} + G(q) = \Omega\tau + \phi_{diss}(\dot{q}, q, t) + \langle \lambda(t), f(q) \rangle$$

We can propose our **state variables**:

$$\begin{cases} q_a = q \\ q_b = \dot{q} \end{cases}$$

Then, the *dynamics* of the proposed **state variables** are:

$$\dot{q}_a = \dot{q} = q_b$$

$$\dot{q}_b = \ddot{q} = B^{-1}(q_a) [\Omega\tau + \phi_{diss}(q_b, q_a, t) + \langle \lambda(t), f(q_a) \rangle]$$

$$\begin{cases} \dot{q}_a = q_b \\ \dot{q}_b = f(q_a, q_b) + g(q_a)(\Omega\tau + \langle \lambda(t), f(q_a) \rangle) \end{cases}$$

Now, we need to propose a controller so we can control our dynamics disregarding the *system*, in

$$\dot{q}_b = f(q_a, q_b) + g(q_a)(\Omega\tau + \langle \lambda(t), f(q_a) \rangle) = -K_P q_a - K_D q_b$$

To do this, we may simply propose a *PD controller* for the *torque* τ that takes into account

$$\tau = \Omega^{-1} [g^{-1}(q_a) (-K_P q_a - K_D q_b - f(q_a, q_b)) - \langle \lambda(t), f(q_a) \rangle]$$

$$\tau = (J^T)^{-1} [B^{-1}(q_a) (-K_P q_a - K_D q_b - f(q_a, q_b))]$$

When substituting into the dynamics we get :

$$\dot{q}_b = f(q_a, q_b) + g(q_a) (\cancel{\Omega \Omega^{-1}} [g^{-1}(q_a) (-K_P q_a - K_D q_b - f(q_a, q_b))] \cancel{- \langle \lambda(t), f(q_a) \rangle} + \langle \lambda(t), f(q_a) \rangle)$$

$$\dot{q}_b = f(q_a, q_b) + \cancel{g(q_a)[g^{-1}(q_a)]}(-K_P q_a - K_D \dot{q}_b - f(q_a, \dot{q}_b))$$

$$\dot{q}_b = \cancel{f(q_a, q_b) - f(q_a, \dot{q}_b)} - K_P q_a - K_D \dot{q}_b$$

$$\dot{q}_b = -K_P q_a - K_D \dot{q}_b$$

We basically made a *controller* for the *system's* acceleration by proposing one for the

$\begin{cases}$

$$\dot{q}_a = q_b \setminus$$

$$\dot{q}_b = -K_P q_a - K_D \dot{q}_b$$

$\end{cases}$

In *vector form* :

$$\frac{d}{dt} \mathbf{q} = \begin{bmatrix} q_a \setminus q_b \end{bmatrix} =$$

$$\begin{bmatrix}$$

$$q_b \setminus$$

$$-K_P q_a - K_D \dot{q}_b$$

$$\end{bmatrix}$$

$$= \begin{bmatrix}$$

$$0_{n \times n} \& I_{n \times n} \setminus$$

$$-K_P \& -K_D$$

$$\end{bmatrix} \begin{bmatrix}$$

$$q_a \setminus$$

$$q_b$$

$$\end{bmatrix}$$

$$\frac{d}{dt} \mathbf{q} = \mathbf{A} \mathbf{q}$$

From this, is obvious to note that $\mathbf{q}$ is a 2×1 vector, and both K_P and K_D are

$\begin{cases}$

$$\dot{q}_a = q_b \setminus$$

$$\dot{q}_b = f(q_a, q_b) + g(q_a)(\Omega + \langle \lambda(t), f(q_a) \rangle) +$$

$$\psi(q_a, \dot{q}_b, t)$$

$\end{cases}$

Where: $\psi(q_a, q_b, \dots)$: Non-modelled sections of the robot. $\psi(\dots, t)$: Effect of all external perturbations

$$\|\Psi\| = \|\psi\|_{Q\{a\}} \times \|TQ\{a\}\| \times \|\mathbf{R}^+\| \times \|\mathbf{R}^n\| \times \|\text{perturbations}\|$$

$$\|\Psi\|^2 \leq \|\psi\|_{Q\{a\}}^2 + \|\psi\|_{Q\{a\}}^2 + \|\psi\|_{Q\{b\}}^2 + \|\psi\|_{Q\{b\}}^2$$

Where: $Q_a \subset \mathbb{R}^n$: Configuration space (positions of q_a) TQ_a : Tangent bundle of Q_a , represents velocities

$$\tau = \Omega^{-1}[g^{-1}(q_a)(-K\{P\}q_a - K\{D\}q_b - f(q_a, q_b))] - \lambda$$

$$\lambda(t, f(q_a)) \angle$$

So we end up with :

$$\dot{q}_b = -K\{P\}q_a - K\{D\}q_b + \psi(q_a, q_b, t)$$

$$\frac{d}{dt} q = \underbrace{\begin{bmatrix} 0_{n \times n} & I_{n \times n} \\ -K\{P\} & -K\{D\} \end{bmatrix} A}_{B} \psi$$

$$q_a \setminus$$

$$q_b$$

$$\end{bmatrix} + \underbrace{\begin{bmatrix} 0 & I \end{bmatrix} B}_{\psi}$$

Where we said our perturbations ψ are bounded by :

$$\|\Psi\| \leq \|\psi\|_{Q\{a\}} \times \|TQ\{a\}\| \times \|\mathbf{R}^+\| \times \|\mathbf{R}^n\| \times \|\text{perturbations}\|$$

$$\|\Psi\|_{Q\{b\}} \leq \|\psi\|_{Q\{b\}} \times \|\mathbf{R}^+\| \times \|\mathbf{R}^n\| \times \|\text{perturbations}\|$$